

Computer Systems 2

Computing and Mathematics

May, 2026

Computer Systems 2 (A11062)

Short Title:	Computer Systems 2	
Faculty	Science and Computing	
Department	Computing and Mathematics	
Cluster	Networks and Cloud	
Author	EREADE	
Credits	5	Level: Introductory

Description of Module / Aims

This module follows on from Computer Systems 1 and focuses mostly on operating systems. Topics covered include core operating system functions such as process, memory and file management as well as an introduction to distributed systems and virtualisation. As with Computer Systems 1, there is an emphasis on practical, lab-based exercises with command line tools and some scripting.

Programmes

COMP-0546	BSc (Hons) in Applied Computing (International) (WD_KACMI_B)	1 / 2 / M
COMP-0546	BSc (Hons) in Applied Computing (WD_KACCM_B)	1 / 2 / M
COMP-0546	BSc (Hons) in Applied Computing (WD_KCOMP_B)	1 / 2 / M
COMP-0546	BSc (Hons) in Computer Forensics and Security (WD_KCOFO_B)	1 / 2 / M
COMP-0546	BSc (Hons) in Computer Science (WD_KCMSC_B)	1 / 2 / M
COMP-0546	BSc (Hons) in Creative Computing (WD_KCRCO_B)	1 / 2 / M
COMP-0546	BSc (Hons) in Software Engineering (WD_KDEVP_BI)	1 / 2 / M
COMP-0546	BSc (Hons) in Software Systems Development (WD_KDEVP_B)	1 / 2 / M
COMP-0546	BSc (Hons) in the Internet of Things (International) (WD_KINTT_BI)	1 / 2 / M
COMP-0546	BSc in Applied Computing (WD_KCOMP_D)	1 / 2 / M
COMP-0546	BSc in Information Technology (WD_KINFT_D)	1 / 2 / M
COMP-0546	BSc in Multimedia Applications Development (WD_KMULA_D)	1 / 2 / M
COMP-0546	BSc in Software Systems Development (WD_KCOMC_D)	1 / 2 / M

Indicative Content

- Operating system structure: components, services and utilities
- Process Management
- File management
- Memory management
- Virtualisation
- Basic concepts of networks and distributed systems

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the memory management, process management and file management components of a modern operating system.
2. Explain basic concepts and theory of networked operating systems and virtualisation.
3. Configure a contemporary operating system (within a virtual machine environment), integrating the services necessary to support basic two/three tier applications, and demonstrating their operation with a small application.
4. Demonstrate competency in a limited set of utilities provided by a contemporary operating system.
5. Complete basic automation tasks using scripting.

Learning and Teaching Methods

- There is a strong emphasis on practical, lab-based exercises.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	50%	
<i>Assignment</i>	50%	3,4,5
Final Written Examination	50%	1,2,4

Assessment Criteria

<40%: Unable to describe operating system concepts. Unable to use command line utilities of an OS.

Unable to write a basic effective script to demonstrate automation of a configuration task.

40%-49%: Can describe operating system concepts. Can configure an OS and deploy an application. Can use basic command line utilities and write a script to demonstrate automation.

50%-59%: All of the above. Can describe concepts in more depth. Additionally can write a script to automate a non-trivial task.

60%-69%: In addition, can carry out more advanced configuration and scripting tasks.

70%-100%: All the above to an excellent level. In addition can research and execute previously unseen configuration and scripting tasks independently.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	12
Practical	24	12
Independent Learning	87	111

Supplementary Material

- Forouzan, B. *_Foundations of Computer Science_*. New York: Cengage Learning, 2013.
- Siberschatz, A., P. Galvin and G. Gagne. *_Operating Systems Concepts_*. 9th Ed. NY: Wiley, 2013.
- Stallings, W. *_Operating Systems: Internals and Design Principles_*. 8th Ed. NY: Pearson, 2014.

Requested Resources

- COMPUTER LAB: BYOD Lab
- Room Type: Computer Lab